

# MOHAMMED AMIR KHABAZA

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## EDUCATION

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**Georgia Tech** — *M.S. in Cybersecurity - (Information Security) - Online*

*Incoming: August 2026*

**UC Merced** — *B.S. in Computer Science & Engineering*

*Graduated: May 2026*

GPA: 3.83

**Relevant Coursework:** Data Structures & Algorithms, Linear Algebra, Probability & Statistics, Machine Learning, Discrete Math, Boolean Algebra

## SKILLS

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**Programming:** Python, C++, Java, JavaScript/TypeScript, SQL, Ruby, Go, MATLAB, Kotlin

**AI & Data Tools:** TensorFlow, Keras, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, MongoDB, Tableau

**Systems & Platforms:** React, Node.js, Flask, FastAPI, Git, Perforce, Linux, CUDA, OpenCV, MediaPipe

**Specialties:** Data Analysis, Model Evaluation, Computer Vision, Agentic AI, Human-Computer Interaction

## EXPERIENCE

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**NASA Ames Research Center – Mountain View, CA**

*Software Engineer Intern*

*May 2026 – Present*

- Engineered a cross-platform **PsychoPy** frontend for **9-point eye-tracker calibrations**, logging VSYNC-aligned timestamps for precise gaze telemetry.
- Developed a Python pipeline (**NumPy, Pandas**) to synchronize hardware clocks and **correct physical tracking drift** using **2D affine transformations**.
- Computed kinematic features and integrated LLMs to analyze oculomotor behavior and mental workload for a lunar-lander analog tracking study.

**PADSYS – Merced, CA**

*Undergraduate Research Intern*

*August 2025 – Present*

- Investigating I/O bottlenecks in High-Performance Computing (HPC) systems by profiling AI workloads, focusing on processes like **checkpointing** and **data loading**.
- Deploying and benchmarking computer vision models (**YOLOv12, mmdetection library**) in containerized Ubuntu environments using **Docker** to analyze **I/O performance**.

## PROJECTS

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**AIVue: Control with Vision (Winner, SF Hacks 2025)**

*Team Lead & Developer*

*March 2025 - Present*

- Engineered a **hands-free computer control system** for people with disabilities using **eye-tracking, blink detection, and speech commands**.
- Implemented computer vision with **Python, OpenCV, MediaPipe**, and Kalman filters; integrated **speech-to-text AI** and **MongoDB Atlas**.
- Improved gaze accuracy by 32% and reduced latency by 18% through adaptive AI filtering.

Devpost: <https://devpost.com/software/aivue-control-with-vision>

**EthicScope (Winner, HackDavis 2025)**

*Founding Developer*

*April 2025 - May 2026*

- Designed a mobile and web app to analyze product barcodes and assess ethical & environmental impact using ML-based data aggregation.
- Built backend with **Flask, MongoDB, and Cerebras**; developed visual data dashboards using **React + Tailwind**.
- Helped over 1,000 simulated test users make informed, transparent purchasing decisions.

Devpost: <https://devpost.com/software/ethicscope>

**AI Tumor Detection Machine Learning Model: AI4ALL**

*Undergraduate Machine Learning Researcher*

*January 2024 - October 2025*

- Developed a **VGG-16 CNN** for brain MRI tumor classification, achieving **94% accuracy** on validation data.
- Collaborated with interdisciplinary researchers to improve model interpretability and bias detection.
- Used **TensorFlow, Keras, and Scikit-learn** for model training, preprocessing, and ROC analysis.